

Problem D: Dynamic Sports Business Decision Model (WNBA)

Example Team: **Indiana Fever** (Indianapolis)

Team 2633018

Summary Sheet

Problem. A professional sports team is an entertainment business. The owner must maximize profit and long-run franchise value while managing competitive performance. Required decisions include: (i) leverage adjustment over time under changing performance and economic conditions, (ii) player acquisition using standard league mechanisms (draft/free agency/trade/waivers), (iii) adaptation during league expansion, (iv) an additional business decision (we choose ticket pricing), and (v) adjustment when a key player is injured.

Team and League (Data Used). We use public, team-specific data for the **Indiana Fever** and WNBA constraints.

- Fever arena capacity (basketball): 18,274.
- Fever average home attendance: 16,560.23.
- Fever home games: 22.
- Fever market size (metro pop proxy): 2,080,000.
- Season length: 44 games.
- Salary cap (2025): \$1,507,100.
- Roster size: 11–12.

Our Key Idea. A model is dynamic only if it specifies: (a) a state that evolves over time, and (b) explicit decision rules (policies) mapping state $\rightarrow$ actions. We define the state

$$X_t = (Q_t, B_t, C_t, D_t, M_t, E_t)$$

and management decisions

$$u_t = (\Delta D_t, m_t, p_t, a_t) = \pi(X_t),$$

where ΔD_t is borrowing/paydown, m_t marketing spend, p_t ticket pricing, and a_t roster actions.

How we measure players (from our Excel). For each player i we build:

$$(b_i, \pi_i, s_i, c_i),$$

where:

- b_i = **performance score** computed from per-game box score stats in our data.
- π_i = **injury risk** computed from games missed in the season.
- s_i = **popularity score** computed from social media followers.
- c_i = salary.

Core System. Performance and popularity build demand; demand drives revenue; revenue changes cash; cash and debt constrain future actions. This creates feedback loops.

- Team quality:

$$Q_t = \sum_{i \in S_t} b_i(1 - \pi_i).$$

- Brand/demand stock:

$$B_{t+1} = (1 - \rho)B_t + \alpha \text{Wins}_t + \beta \sum_{i \in S_t} s_i(1 - \pi_i) + \gamma m_t.$$

- Attendance and ticket pricing:

$$\text{Att}_g = \min\left(18,274, A_0 e^{\xi B_t + \zeta Q_t} p_g^{-\epsilon}\right).$$

- Profit and financing:

$$\Pi_t = R_t - W_t - O_t - m_t - r_t D_t, \quad C_{t+1} = C_t + \Pi_t + \Delta D_t.$$

Dynamic Decision Rules.

- Leverage rule: borrow only when cash is healthy and profit outlook is positive; deleverage under stress or low cash.
- Roster rule: choose keep/acquire decisions under cap/roster constraints using a value metric combining performance, popularity, and injury risk.
- Pricing rule: set game-level price to maximize expected revenue given current Q_t, B_t .
- Expansion rule: adjust salary pressure and demand substitution based on location overlap and re-optimize.

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1 Introduction and Problem Restatement

We build a decision system for a professional sports team where the owner aims to maximize profit and franchise value while remaining competitively viable. The prompt requires a dynamic model that adapts decisions over time to changing performance, injuries, economic conditions, and league expansion. We choose the Indiana Fever as an example WNBA team and use publicly obtainable data for salaries, box score performance, games played, and popularity proxies.

2 Data Inputs Used

Our Excel workbook provides four tables:

- **League table:** season_year, salary_cap, roster_bounds, total_games, payroll minimum, player max/supermax.
- **Teams table:** city, arena capacity, average attendance, home games, market size, latitude/longitude, win%.
- **Raw player stats:** per-game PTS/REB/AST/STL/BLK/TO/MIN and games_played.
- **Players table:** player name, salary, position, age, social media followers.

3 Assumptions

3.1 Business/Finance Assumptions

- (A1) The owner can adjust debt by ΔD_t (borrow or pay down) and chooses marketing spend m_t .
- (A2) Higher leverage increases financing cost and risk (interest rate depends on leverage and macro conditions).
- (A3) The owner enforces a minimum cash buffer $C_{\min}$ to reduce distress risk.

3.2 Performance/Demand Assumptions

- (A4) Expected team quality is an additive function of expected player contributions, adjusted for injury risk.
- (A5) Demand responds to brand stock B_t and quality Q_t ; ticket demand has price elasticity.
- (A6) Popularity is represented by a measurable proxy (social following) so demand can grow even without wins.
- (A7) Games missed reflect both injury and other unavailability; thus injury risk represents overall availability risk rather than purely medical injury probability

3.3 League/Mechanism Assumptions

- (A8) Team must satisfy salary cap and roster size constraints.
- (A9) Expansion affects salaries (inflation) and local demand (substitution based on location overlap).

4 Notation and Variables

4.1 Time and Sets

- $t = 1, \dots, T$: decision periods (e.g., weeks).
- $g \in \mathcal{G}(t)$: home games in period t (counted in revenue).
- S_t : roster in period t (set of active players).

4.2 State Variables

$$X_t = (Q_t, B_t, C_t, D_t, M_t, E_t)$$

- Q_t : expected on-court quality (computed from roster).
- B_t : brand/demand stock (latent index updated each period).
- C_t : cash buffer (liquidity available at time t).
- D_t : outstanding debt balance at time t .
- $M_t \in \{0, 1\}$: macro stress indicator (scenario variable).
- $E_t \in \{0, 1\}$: expansion environment indicator (scenario variable).

4.3 Decision Variables

$$u_t = (\Delta D_t, m_t, p_t, a_t) = \pi(X_t)$$

- ΔD_t : debt adjustment (borrow if > 0 , pay down if < 0).
- m_t : marketing spend during period t .
- p_t (or p_g): ticket price decision (we treat as game-level).
- a_t : roster actions (keep/sign/trade/waive/draft).

5 How We Construct Player Inputs From Excel

Each player i is mapped into:

$$(b_i, \pi_i, s_i, c_i)$$

where c_i is salary directly from the players table and the other quantities are computed as follows.

5.1 Injury Risk π_i

We define injury risk as the fraction of games missed:

$$\pi_i = \frac{G_{\text{total}} - G_i}{G_{\text{total}}}, \quad (1)$$

where $G_{\text{total}} = 44$ (league total games) and G_i is games_played from raw stats.

5.2 Popularity Score s_i

Let F_i be social_media_followers. Followers are heavy-tailed, so we use a log transform:

$$s_i = \log(1 + F_i). \quad (2)$$

5.3 Performance Score b_i

We use the exact per-game formula implemented in our dataset:

$$b_i = \text{PTS}_i + 1.2 \text{REB}_i + 1.5 \text{AST}_i + 3(\text{STL}_i + \text{BLK}_i) - 2 \text{TOV}_i. \quad (3)$$

This yields a transparent impact proxy from standard box score contributions.

6 Dynamic System Model

6.1 Team Quality (Roster $\rightarrow$ Performance)

$$Q_t = \sum_{i \in S_t} b_i(1 - \pi_i). \quad (4)$$

6.2 Brand/Demand Stock (Performance + Popularity + Marketing)

$$B_{t+1} = (1 - \rho)B_t + \alpha \text{wins}_t + \beta \sum_{i \in S_t} s_i(1 - \pi_i) + \gamma m_t. \quad (5)$$

Marketing spending follows a capped rule to reflect diminishing returns:

$$m_t = \min(m_{\max}, \kappa(C_t - C_{\min})^+), \quad (6)$$

where $(x)^+ = \max(x, 0)$.

6.3 Attendance and Ticket Demand (Pricing Decision)

For each home game g :

$$\text{Att}_g = \min(\text{Cap}, A_0 e^{\xi B_t + \zeta Q_t} p_g^{-\varepsilon}), \quad (7)$$

where for Indiana Fever $\text{Cap} = 18,274$.

Rationale. The term $p_g^{-\epsilon}$ represents constant-elasticity ticket demand, where $\epsilon > 0$ controls consumer sensitivity to price. The exponential factor $e^{\xi B_t + \zeta Q_t}$ captures increased willingness to attend when the team is popular or successful. The capacity constraint $\min(\text{Cap}, \cdot)$ enforces the physical limit of the arena.

6.4 Revenue and Profit

We model two revenue streams:

(1) Game-day revenue: ticket price times attendance, plus optional per-attendee ancillary spend.

- p_g = ticket price for game g
- Att_g = attendance at game g
- a = ancillary revenue per attendee (concessions/parking/merch inside venue), treated as a scenario parameter

(2) Brand-linked revenue: sponsorship/media revenue that scales with brand.

- μ = brand-to-revenue scaling parameter (scenario parameter)

Thus total revenue in period t :

$$R_t = \sum_{g \in t} (p_g + a) \text{Att}_g + \mu B_t. \quad (8)$$

Total costs:

- W_t = payroll paid in period t (from selected roster salaries; can be amortized weekly)
- O_t = other operating costs (travel, staff, arena ops), treated as scenario parameter
- m_t = marketing spend decision
- $r_t D_t$ = interest expense on debt

Profit:

$$\Pi_t = R_t - W_t - O_t - m_t - r_t D_t. \quad (9)$$

6.5 Cash and Debt Updates

$$C_{t+1} = C_t + \Pi_t + \Delta D_t, \quad D_{t+1} = D_t + \Delta D_t. \quad (10)$$

6.6 Financing Cost (Macro + Leverage)

$$r_t = r_0 + r_M M_t + r_L \frac{D_t}{V_t}. \quad (11)$$

(r_0, r_M, r_L and franchise value V_t are scenario parameters estimated/calibrated in simulation.)

7 Decision Rules

7.1 Owner Leverage Policy $\Delta D_t = \pi_D(X_t)$

$$\Delta D_t = \begin{cases} +\Delta^+ & C_t \geq C_{\text{target}}, \hat{\Pi}_{t:t+H} > 0, M_t = 0 \\ 0 & C_{\min} \leq C_t < C_{\text{target}} \\ -\Delta^- & C_t < C_{\min} \text{ or } M_t = 1 \end{cases} \quad (12)$$

Interpretation. When cash reserves exceed a target level and short-term profit outlook is positive under stable macro conditions, the owner borrows to finance growth investments. When liquidity is moderate, borrowing is paused. When cash falls below a safety threshold or macro stress occurs, debt is reduced to limit financial risk.

7.2 Ticket Pricing Policy $p_g = \pi_p(X_t)$

$$p_g^* \in \arg \max_{p_{\min} \leq p \leq p_{\max}} (p + a) \cdot \text{Att}_g(p; Q_t, B_t). \quad (13)$$

Interpretation. For each home game, the team selects the ticket price that maximizes expected per-game revenue given current team quality and brand demand. As performance improves or star popularity increases, the demand curve shifts outward and higher optimal prices are recommended. After injuries or performance declines, optimal prices decrease automatically.

7.3 GM Roster Acquisition Policy $a_t = \pi_a(X_t)$

$$\text{Score}_i = \frac{\omega_b b_i + \omega_s s_i - \omega_r \pi_i}{c_i}. \quad (14)$$

$$\max \sum_i y_i (\omega_b b_i + \omega_s s_i - \omega_r \pi_i) \quad (15)$$

subject to:

$$\sum_i y_i c_i \leq 1,507,100 \quad (2025 \text{ cap}) \quad (16)$$

$$11 \leq \sum_i y_i \leq 12 \quad (\text{roster size}). \quad (17)$$

Where draft/FA/trade/waivers enter. The acquisition *method* is represented by constraints on the candidate set:

- Draft: candidates restricted to draft pool and number of picks owned.
- Free agency: candidates restricted to free agents; only salary cap applies.
- Trades: impose an asset budget and allowable incoming/outgoing salaries.
- Waivers: allow in-season replacement from waiver pool (small candidate set).

8 Expansion and Location Effects

Let $E_t \in \{0, 1\}$ indicate expansion environment.

8.1 Salary Inflation (Talent Competition)

$$c_i \leftarrow c_i(1 + \delta_c E_t). \quad (18)$$

8.2 Demand Substitution (Location Overlap)

$$\text{Overlap}(d) = e^{-d/\tau}, \quad (19)$$

$$B_{t+1} \leftarrow B_{t+1} - \delta_{\text{sub}} \text{Overlap}(d) E_t + \delta_{\text{league}} E_t. \quad (20)$$

9 How the Model Adjusts After a Key Injury

If a key player k is injured, then π_k increases, lowering Q_t in (4). This reduces attendance (7), revenue (8), and cash (10). The policies respond:

- Recompute optimal ticket prices using (13).
- Re-run roster selection with waiver/free-agent pools (if enabled).
- Leverage policy (12) shifts toward deleveraging if C_t approaches $C_{\min}$ or $M_t = 1$.

10 Simulation Implementation and Experimental Design

This section explains how the dynamic model is executed as a forward simulation over a season. Each simulated week updates the roster decision, ticket price, revenue and costs, leverage action, and brand state. The simulation is implemented in Python and reads all inputs (players, raw stats, league limits, team capacity) from a single Excel workbook.

10.1 Inputs and Derived Player Metrics

We construct a unified player table by merging two sheets: (i) player attributes (salary, team, and social media followers) and (ii) per-game box score statistics. From these, the model computes the following quantities used in roster quality and brand:

Performance score. For each player i , a composite performance score b_i is computed from per-game statistics:

$$b_i = \text{PTS}_i + 1.2 \text{REB}_i + 1.5 \text{AST}_i + 3(\text{STL}_i + \text{BLK}_i) - 2 \text{TO}_i. \quad (21)$$

Injury risk proxy. Let G be the league total games (from the league sheet). Let G_i be games played by player i . The injury-risk proxy is

$$\pi_i = \frac{G - G_i}{G}, \quad \pi_i \in [0, 1]. \quad (22)$$

Popularity score. Popularity is proxied by social media followers using a log transform:

$$s_i = \log(1 + \text{followers}_i). \quad (23)$$

10.2 Weekly Simulation Loop

We simulate $T = 11$ weekly decision periods. The state at week t is cash C_t , debt D_t , and brand stock B_t . Each week proceeds in the following order (matching the code implementation):

1. **Roster selection (heuristic optimizer).** From a candidate pool, the algorithm selects a roster subject to the salary cap and roster size limits (minimum/maximum roster size). Each candidate player has an acquisition channel (keep/free agent/trade/draft/waiver). Trade and draft selections are additionally constrained by an asset budget and number of draft picks available.

The roster objective maximizes total owner value:

$$\text{Value}(i) = w_b b_i + w_s s_i - w_r \pi_i, \quad (24)$$

and the heuristic ranks players by value-per-dollar to build a feasible roster under the constraints.

2. **Injury trigger and waiver option.** If a designated key player experiences an injury shock in week t (scenario input), their risk is increased by +0.25 and their performance is reduced (multiplied by 0.6). If the key player's risk exceeds the threshold $\pi_i \geq \pi_{\text{th}}$, the model re-optimizes the roster and allows waiver candidates in the selection set for that week.
3. **Team quality computation.** Given the chosen roster, team quality is computed as:

$$Q_t = \sum_{i \in \mathcal{R}_t} b_i(1 - \pi_i) - \eta \cdot \text{Imbalance}(\mathcal{R}_t), \quad (25)$$

where the imbalance term is optional and set to $\eta = 0$ in our baseline runs.

4. **Marketing decision.** Marketing spend is a cash-based rule with an upper bound:

$$m_t = \min(200,000, 0.02 \cdot \max(C_t - C_{\min}, 0)). \quad (26)$$

5. **Ticket pricing by grid search.** Given (B_t, Q_t) , the model selects ticket price p_t by maximizing per-home-game ticket-plus-ancillary revenue over a discrete price grid:

$$p \in \{p_{\min}, p_{\min} + \Delta p, \dots, p_{\max}\}.$$

Attendance for a given price is:

$$\text{Att}_t(p) = \min(\text{Cap}, A_0 \exp(\xi B_t + \zeta Q_t) p^{-\varepsilon}), \quad (27)$$

so the chosen price is

$$p_t^* = \arg \max_p (p + a) \text{Att}_t(p). \quad (28)$$

6. **Weekly revenue and costs.** Let h be home games per week (home games divided by T). Weekly ticket revenue is $h \cdot (p_t^* + a) \text{Att}_t(p_t^*)$. Sponsorship/media revenue is proportional to brand:

$$R_t = h(p_t^* + a) \text{Att}_t(p_t^*) + \mu B_t. \quad (29)$$

Payroll is approximated as season payroll spread across T :

$$W_t = \frac{\sum_{i \in \mathcal{R}_t} \text{salary}_i}{T}. \quad (30)$$

Operations cost is similarly spread across periods: $O_t = O_t/T$.

The interest rate follows:

$$r_t = r_0 + r_M M_t + r_L \left(\frac{D_t}{V} \right), \quad (31)$$

where $M_t \in \{0, 1\}$ is the macro stress indicator and V is franchise value (scenario input). Profit is:

$$\Pi_t = R_t - W_t - O_t - m_t - r_t D_t. \quad (32)$$

7. **Leverage policy and state transition.** Using a k -week moving average forecast $\hat{\Pi}_t$ (with $k = 3$), the debt change ΔD_t follows:

$$\Delta D_t = \begin{cases} -d_-, & C_t < C_{\min} \text{ or } M_t = 1, \\ +d_+, & C_t \geq C_{\text{target}} \text{ and } \hat{\Pi}_t > 0 \text{ and } M_t = 0, \\ 0, & \text{otherwise.} \end{cases} \quad (33)$$

Cash and debt update as:

$$C_{t+1} = C_t + \Pi_t + \Delta D_t, \quad D_{t+1} = D_t + \Delta D_t.$$

8. **Wins sampling and brand update.** Expected win probability is a logistic function of Q_t and weekly wins are sampled from a binomial distribution. Brand stock updates as:

$$B_{t+1} = (1 - \rho) B_t + \alpha \text{wins}_t + \beta \sum_{i \in \mathcal{R}_t} s_i (1 - \pi_i) + \gamma m_t, \quad (34)$$

with an optional expansion adjustment (described in the scenarios below).

10.3 Parameter Choices and Ranges

All parameters are stored in a single configuration object. To keep the simulation interpretable and numerically stable, we use bounded ranges consistent with the implemented mechanisms:

Ticket price grid. We search prices from \$5 to \$250 in increments of \$2:

$$p_{\min} = 5, \quad p_{\max} = 250, \quad \Delta p = 2.$$

Demand sensitivities and elasticity. We use $(\xi, \zeta) = (0.02, 0.02)$ to allow brand and quality to shift demand smoothly without dominating the capacity cap, and $\varepsilon = 1.2$ to represent elastic demand where higher prices reduce attendance.

Calibration of baseline demand scale A_0 . Rather than choosing A_0 arbitrarily, we calibrate it using a reference price $p_{\text{ref}} = 25$ and a target attendance from the team sheet (capped by arena capacity). Solving the attendance equation at the reference state (B_0, Q_{ref}) gives:

$$A_0 = \text{Att}_{\text{target}} \cdot p_{\text{ref}}^\varepsilon \cdot \exp(-\xi B_0 - \zeta Q_{\text{ref}}). \quad (35)$$

Finance parameters and policy thresholds. We use baseline interest $r_0 = 0.05$, macro add-on $r_M = 0.03$, and leverage premium $r_L = 0.02 \cdot (D/V)$. Liquidity thresholds and step sizes are:

$$C_{\min} = 200,000, \quad C_{\text{target}} = 500,000, \quad d_+ = d_- = 50,000, \quad k = 3.$$

Roster objective weights. The roster optimizer scores each player with weights $(w_b, w_s, w_r) = (1.0, 0.4, 1.0)$ to reward on-court performance, partially reward popularity, and penalize injury risk.

Expansion parameters. If expansion is active ($E_t = 1$), candidate salaries are inflated by $\delta_c = 0.08$. Brand is adjusted by:

$$B_{t+1} \leftarrow B_{t+1} + \delta_{\text{league}} - \delta_{\text{sub}} \exp(-d/\tau),$$

with $\delta_{\text{league}} = 0.10$, $\delta_{\text{sub}} = 0.12$, and $\tau = 800$ km, where d is the great-circle distance between markets.

11 Scenario Runs and Reported Outputs

We run four scenarios over $T = 11$ weeks using the same initial state (C_0, D_0, B_0, V) but different exogenous inputs:

- **Baseline:** $M_t = 0$ for all t , $E_t = 0$.
- **Macro stress:** $M_t = 1$ for all t (higher interest and conservative leverage).
- **Injury shock:** one-week shock at week 5 for the key player, activating the waiver option and reducing that player's contribution.
- **Expansion:** $E_t = 1$ with an expansion market location, triggering salary inflation and a distance-based brand adjustment.

Recorded outputs. For each run, we record weekly time series of $\{Q_t, B_t, p_t, Att_t, R_t, W_t, m_t, r_t, D_t, \Delta D_t, \Pi_t, C_t\}$ and summarize each scenario by total profit, average profit, final cash/debt, minimum cash, average ticket price, average attendance, average brand, and the number of macro or waiver weeks.

Table 1: Scenario Performance Summary

	scenario	total_profit	avg_profit	final_cash	final_debt	min_cash	avg_price	avg_attendance	avg_brand	avg_quality_Q	weeks_waivers	weeks_macro
3	expansion	7.960276e+07	7.236614e+06	7.007155e+07	700000.0	500000.0	178.272727	18210.255568	137.613074	53.799318	0	0
2	injury	7.956249e+07	7.232954e+06	7.003019e+07	700000.0	500000.0	177.909091	18232.415709	137.255729	53.799318	0	0
1	macro_stress	7.920901e+07	7.200819e+06	6.894685e+07	0.0	500000.0	176.818182	18205.029687	135.750259	53.799318	0	11
0	baseline	5.607451e+07	5.097682e+06	4.670713e+07	700000.0	500000.0	123.181818	18213.201936	105.233829	53.799318	0	0

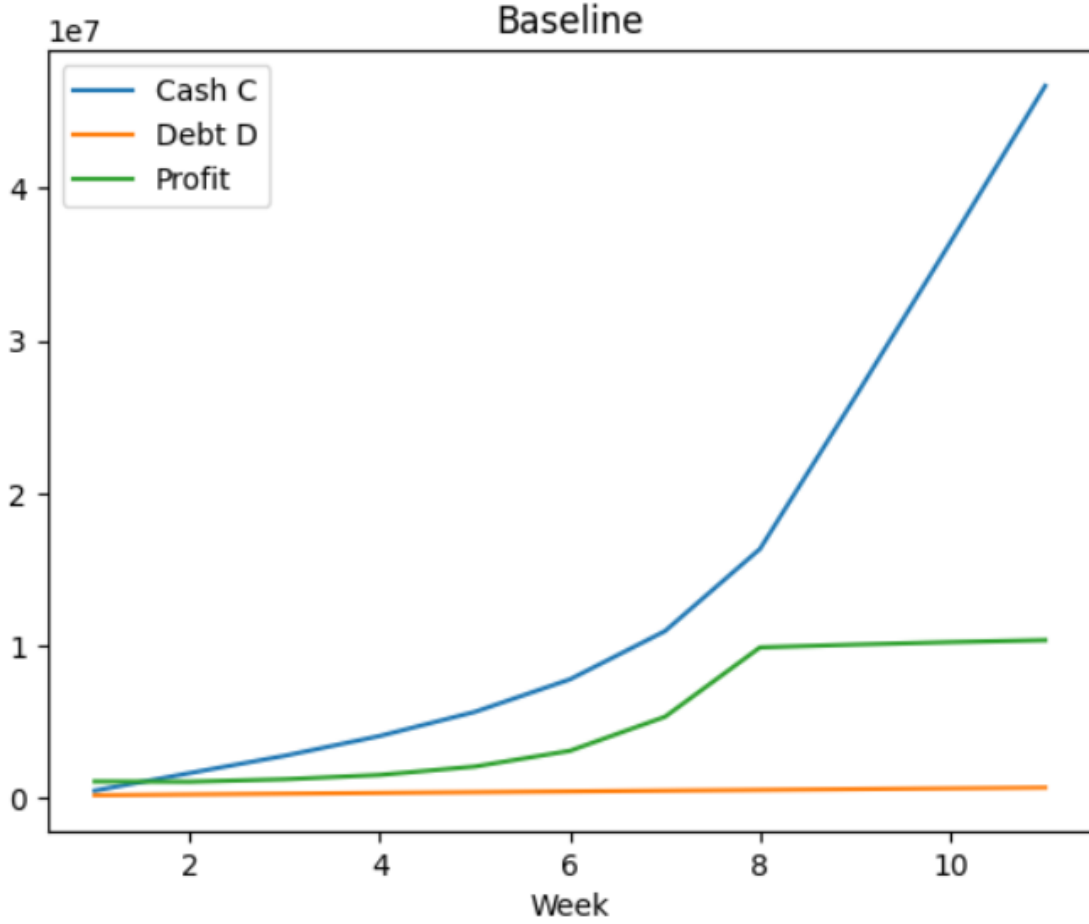


Figure 1: Baseline Scenario: Cash, Debt, and Profit Over Time

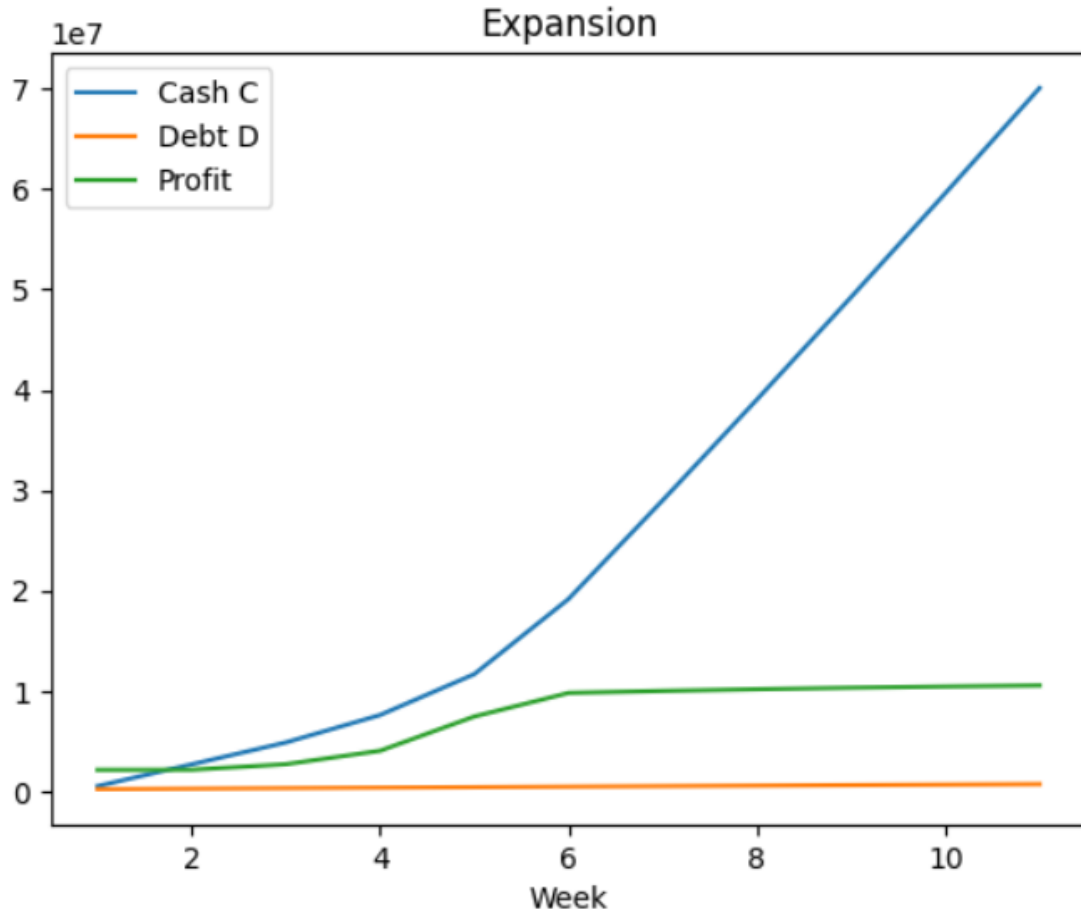


Figure 2: Expansion Scenario: Cash, Debt, and Profit Over Time

Table 2: Simulated Weekly Cash Flow and Profit Dynamics

	week	Q	B	price_p	att_home_game	revenue	payroll_W	marketing_m	r_t	debt_D	dD	profit	cash_C
0	1	53.799318	10.000000	23.0	18274.000000	1.182988e+06	50488.272727	6000.000000	0.05008	200000.0	50000.0	1.102847e+06	5.000000e+05
1	2	53.799318	11.472083	23.0	18274.000000	1.190348e+06	50488.272727	29056.947273	0.05010	250000.0	50000.0	1.084642e+06	1.652847e+06
2	3	53.799318	19.790438	27.0	18274.000000	1.378132e+06	50488.272727	51749.783889	0.05012	300000.0	50000.0	1.247222e+06	2.787489e+06
3	4	53.799318	33.518890	35.0	17700.894112	1.689871e+06	50488.272727	77694.219238	0.05014	350000.0	50000.0	1.530503e+06	4.084711e+06
4	5	53.799318	52.971405	47.0	18274.000000	2.274997e+06	50488.272727	109304.288994	0.05016	400000.0	50000.0	2.081504e+06	5.665214e+06
5	6	53.799318	79.139064	73.0	18244.888841	3.351367e+06	50488.272727	151934.371000	0.05018	450000.0	50000.0	3.112727e+06	7.796719e+06
6	7	53.799318	114.120599	131.0	18207.438345	5.632271e+06	50488.272727	200000.000000	0.05020	500000.0	50000.0	5.343046e+06	1.095945e+07
7	8	53.799318	158.274592	249.0	18274.000000	1.018421e+07	50488.272727	200000.000000	0.05022	550000.0	50000.0	9.892463e+06	1.635249e+07
8	9	53.799318	195.805486	249.0	18274.000000	1.037186e+07	50488.272727	200000.000000	0.05024	600000.0	50000.0	1.007759e+07	2.629496e+07
9	10	53.799318	227.706746	249.0	18274.000000	1.053137e+07	50488.272727	200000.000000	0.05026	650000.0	50000.0	1.023458e+07	3.642255e+07
10	11	53.799318	254.772817	249.0	18274.000000	1.066670e+07	50488.272727	200000.000000	0.05028	700000.0	50000.0	1.036738e+07	4.670713e+07

Total revenue: 58454117.33671294
Total payroll: 555371.0
Total ops cost: 1650000.0
Total profit: 56074505.7263189

11.1 Monte Carlo Robustness Check

Because weekly wins are stochastic, we repeat the simulation $N = 200$ times per scenario with different random seeds. For each trial we record total profit, final cash, final debt, minimum cash, and a bankruptcy indicator $\mathbb{I}\{\min_t C_t < 0\}$. We report descriptive statistics (mean, standard deviation, and quantiles) and the simulated bankruptcy rate.

Table 3: Monte Carlo summary statistics over 200 trials (baseline and macro stress)

	trial	total_profit	final_cash	final_debt	min_cash	bankrupt_flag
count	200.000000	2.000000e+02	2.000000e+02	200.0	200.0	200.0
mean	99.500000	7.958702e+07	7.005406e+07	700000.0	500000.0	0.0
std	57.879185	2.303849e+04	2.281814e+04	0.0	0.0	0.0
min	0.000000	7.952489e+07	6.999295e+07	700000.0	500000.0	0.0
25%	49.750000	7.957191e+07	7.003910e+07	700000.0	500000.0	0.0
50%	99.500000	7.958542e+07	7.005274e+07	700000.0	500000.0	0.0
75%	149.250000	7.960578e+07	7.007276e+07	700000.0	500000.0	0.0
max	199.000000	7.963893e+07	7.010528e+07	700000.0	500000.0	0.0

Bankruptcy rate (baseline): 0.0

Bankruptcy rate (macro): 0.0

Letter to Owner and General Manager

To: Indiana Fever Ownership and General Manager

From: Team 2633018

Subject: Decision System for Leverage, Roster Acquisition, Pricing, Expansion, and Injury Response

Date: July 29, 2026

Dear Owner and General Manager,

We built a dynamic decision system that updates over the season based on observed performance, demand, cash, debt, economic conditions, and league expansion environment. Our model uses measurable public inputs for player performance (box score index), injury risk (games missed), and popularity (social following) to link roster decisions to both wins and revenue.

Leverage. We recommend a state-based leverage rule: borrow only when the cash buffer exceeds a target and near-term profit outlook is positive under stable macro conditions. Under low cash or macro stress, pay down debt and preserve liquidity.

Roster acquisition. We evaluate players by a value-per-dollar score combining performance, popularity, and injury risk, and select players under the WNBA salary cap and roster size constraints. Draft/free agency/trade/waiver mechanisms can be implemented as candidate-set constraints; if not implemented, we still record acquisition channel for reporting.

Pricing. For each home game, we select ticket price to maximize expected revenue given current quality and brand. After an injury shock, quality decreases and the pricing rule adjusts automatically.

Expansion. Expansion increases salary pressure and may change demand through league-wide attention and local substitution. We recommend maintaining higher cash buffers during expansion periods and emphasizing value-per-dollar depth.

Sincerely,
Team 2633018

References

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Wikipedia contributors. (2025, December 4). *2025 Indiana Fever season*. https://en.wikipedia.org/wiki/2025_Indiana_Fever_season

ESPN. (2025). *Indiana Fever 2025 WNBA postseason statistics*. https://www.espn.com/wnba/team/stats/_/name/ind/indiana-fever

OpenAI ChatGPT (February 2nd, 5.2 Version).

Query: Prepare citation in LaTeX for APA style for sources including Her Hoop Stats (2025), Across the Timeline (2024), Wikipedia contributors (2025), and ESPN WNBA statistics (2025).

Output: Generated a formatted APA-style LaTeX References section with italicized titles, wrapped URLs, and clean spacing suitable for ICM/MCM submission.